

JK 12-16 Python First Programmer
Lesson 01 : Variables & Data Types

Duration : 90 mins

Description

Students learn what are variables, data types, concatenation and arithmetic operators and create projects implementing all of the concepts. They create a few simple projects to get them used to writing code.

Objectives

Students will:

1. Recognize live and modern applications of Python.
2. Explain concepts like variables, data types & arithmetic operators with ease.
3. Demonstrate and create simple python programs.
4. Combine different key concepts and create a personal project.
5. Evaluate their level of understanding with the debugging challenge.

Tools

1. Pycharm

Class Structure

Duration	Topic
5 mins	Introduction
5 mins	First Command
10 mins	Variable
10 mins	Data Types
10 mins	Data Types Conversion
15 mins	Concatenation
15 mins	Arithmetic Operators
15 mins	Personal Project
5 mins	Wrap Up

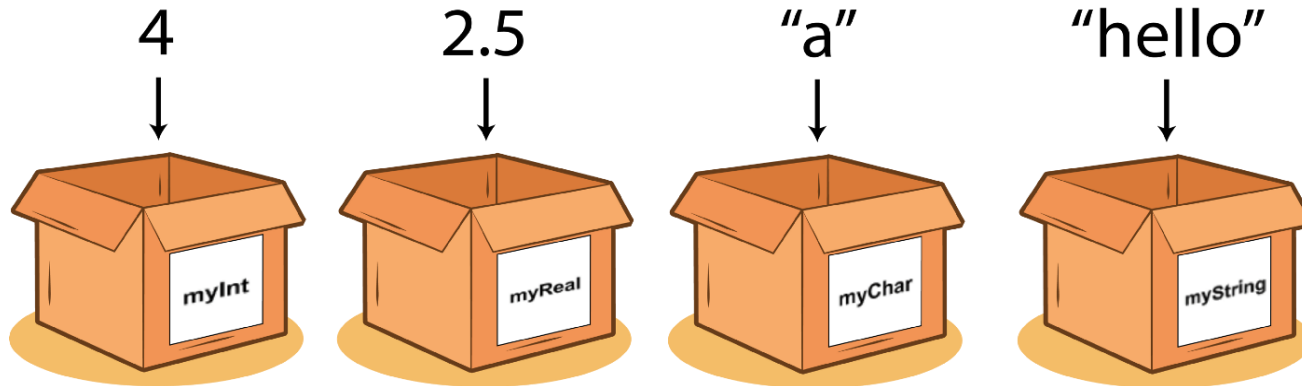
Planned Activities	Resources
Introduction (5 mins)	
Introduction - Welcome students to the class with a brief introduction - Explain about class objectives - Highlight that Python is a good start for new programmers to learn about programming because of the ease of usage.	Website: 10 Famous Software Built With Python
First Command (5 mins)	
First Command – Print - Teacher shows the program used to learn Python that is Pycharm. - Instruct students to open it on their laptops. - Explain Pycharm interface to students and show students how to get up and running with Pycharm. - Demonstrate the command for printing something in the shell using the command <code>Print("text")</code>.	File/Code: 1.Print.py
Individual Work Encourage students to practice the <code>print()</code> command by printing 5 facts about them.	
Variable (10 mins)	

Discuss – Scores

- Show students the game URL in the resources section and let them try the snake game.
- Ask students, what happens after the snake eats the food?
- Explain that the snake will be longer and the length will increase when it eats the food.
- Ask students, have they wondered where we place its lengths? Or when we play a game and there's a score, when can we store the scores?
- Explain the use of variables to store a score or any other value/text/number.

Explain – Variable

- Explain to students that a variable is like a box where you can put a thing inside it. For example, numbers, words, and other information.



- Variable is used to contain something inside.
- When we write the variable's name, it means that we **declare** a variable.
- And when we put a value inside of a variable (it may be number/text/other information), we call it **initialization**.
- To declare a variable, we cannot put a space between the variable name. Also, we cannot put symbols besides underline (_).

Game:

<http://patorjk.com/games/snake/>

File/Code:

[2.Variable.py](#)

- Explain that they also cannot use special symbols like @, !, /, %, and so on. The variable name also cannot use a number as its first letter.
- Some words are reserved and cannot be used as variable names. Show students the reserved keywords in Python. Explain that we cannot use it because the words are used in Python to do their own commands.
- **Python keywords (reserved words):**

Keywords in Python programming language

False	class	finally	is	return
None	continue	for	lambda	try
True	def	from	nonlocal	while
and	del	global	not	with
as	elif	if	or	yield
assert	else	import	pass	
break	except	in	raise	

Explain and Show – Declare & Print

- Show students how to declare variables and initialize them.
- Explain that typing the name of a variable in our program is called calling or referencing.

- To display the value of a variable, we need to call the variable (write the name of the variable) inside the print() function.

Code

- We tell students to create their own variables: their first and last names.
- Tell students to initialize the variables by giving its value with their names.
- Let students print their variables with the print() function.

Data Types (10 mins)

Explain – Data Types

- Explain to students that data types are the types of data that variables can store.
- Four types of data types:
 - **Integer:** a whole number, might be a negative number.
 - **Float:** a decimal number.
 - **String:** text, might be character, words, number as text or sentences.
 - **Boolean:** can only store two types, true or false.
- The difference between integer/float and string is when we store a number as a string data type, we cannot use it with any arithmetic operations. It will display an error message. But, with numeric (integer and float) data types, it can be used for arithmetic operations.
- To initialize a string, we need to add quotation marks at the start and end of the value. But, with numeric and boolean, we don't need to add quotation marks.

Explain and Show

- Show students how to declare variables with different data types.
- Print the value of the variables with the print() function.

File/Code:

[3.DataTypes.py](#)

Code

- In the previous project, instruct students to add more variables: their nickname, age, and Boolean variables.
- Tell students to initialize the variables by giving their values.
- Let students print their variables with the `print()` function.

Data Types Conversion (10 mins)

Explain – Data Types Conversion

- Explain that in most programming languages, we can convert the types of data into other types of data.
- Give an example to students that numeric can be converted into a string and vice versa, but only if the string value is a number.
- Show students how to convert string to integer with **`int()`** function, how to convert integer to string with **`str()`** function, and how to convert integer to float with **`float()`** function.

File/Code:

[4.DataTypesConversion.py](#)

Concatenation (15 mins)

Explain – Concatenation

- Explain to students that concatenation is an act where we can attach variables together in a single `print()` function.
- In Python, to attach those variables in the `print()` function, we use the plus (+) sign.
- Tell students that the best practice in doing concatenation is by using the **f-string concatenation**.

File/Code:

[5.Concatenation.py](#)

Explain and Show – String Concatenation

- Instruct students to declare and initialize 2 variables.
- Show how to concatenate those 2 variables with string.
- Show how to add double quotation marks with the **escape character (\)**. The escape character is used so the double quotation marks inside the print function won't be recognized as the opening or closing of the string inside the print function.

Code

- Instruct students to:
 - Open their previous projects.
 - Concatenate their first and last name variables.
 - Concatenate their nickname and put it inside quotation marks with the escape character.

Arithmetic Operators (15 mins)

Explain – Operators & Arithmetic Operators

- Explain that operators are symbols used to perform a specific type of action.
- There are many types of operators. One of them is the arithmetic operator.
- Arithmetic operators are operators used to perform math calculations.
- Explain that examples of arithmetic operators are:
 - **Addition (+)**: perform an addition between numbers.
 - **Subtraction (-)**: perform a subtraction between numbers.
 - **Multiplication (*)**: perform a multiplication between numbers.
 - **Division (/)**: perform division between numbers.
 - **Modulus (%)**: to divide numbers and return the remainder.
 - **Exponent (**)**: perform exponentially.

File/Code:

[6.ArithmeticOperators.py](#)

Explain and Code – Arithmetic Operators

- Show students the arithmetic operator project.
- Create two number variables and show students how to use the arithmetic operators.

Personal Project (15 mins)

Explain and Show

- Show students the projects.
- Emphasis the requirement to declare two numeric variables as a string with quotation marks.
- Instruct students to print addition, subtraction, multiplication, and division without those variables. Therefore, students need to do the arithmetic operators inside the **print()** function.

Discuss

- Look at students' answers and discuss them together with students.
- Show the codes to students.

File/Code:

[7.PersonalProject.py](#)

Wrap Up (5 mins)

Wrapping Up

- We sum up by asking questions, making sure the students remember the most important things.
- Hands up! Who knows what Variables and Data Types are?
- We thank students for doing a great job overall.

☐ ☆ Good Luck With The Class !! ☆ ☐